

Performance, Propulsion & Climate Engineer

CADEM0016

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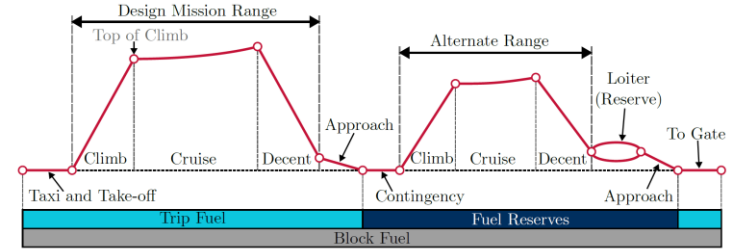
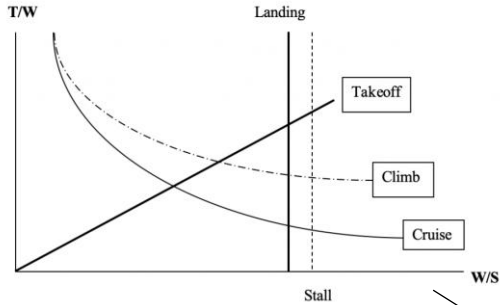


Figure 4: Design mission segment and fuel breakdown

Performance, Propulsion & Climate

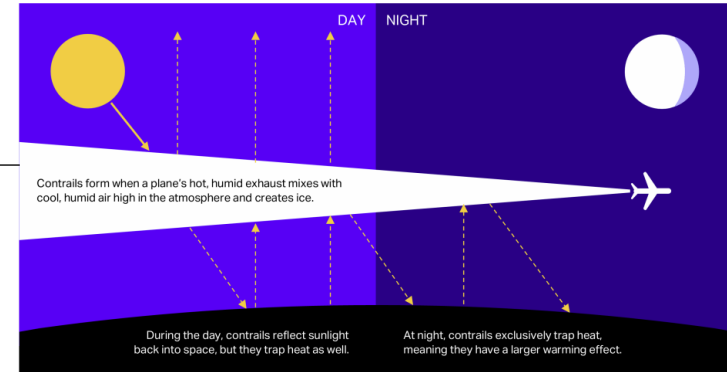
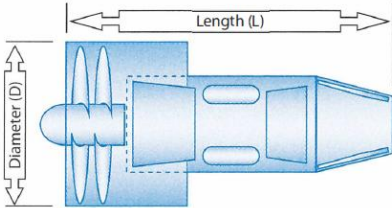


Figure 4 Illustration between day and night contrails. Courtesy of American Airlines

Responsibilities

Sprint	Responsibilities
Development	• Constraint Analysis • Mission Analysis • Climate Model (for cruise) • Propulsion Model • Only consider Design Mission
Refinement	• Climate Model (entire mission) • Mission Model • Consider all flights

Mission Analysis

Mass Fractions

- For a given design mission...
- To estimate fuel mass, we can estimate the change in aircraft **mass**, across each flight segment...

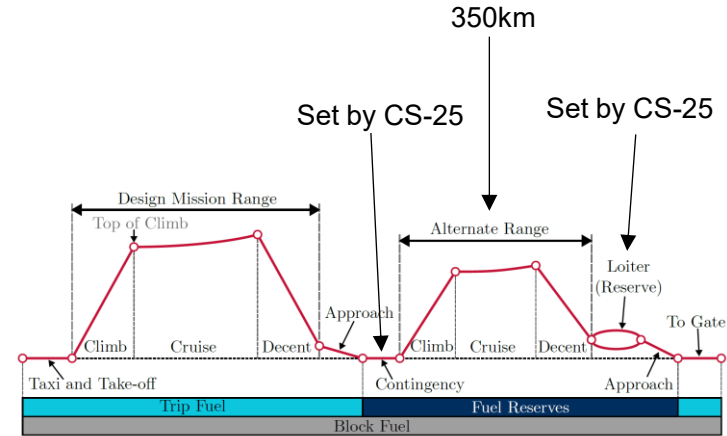


Figure 4: Design mission segment and fuel breakdown

$$\frac{M_f}{M_a} = \frac{M_a - M_e}{M_a} = 1 - \frac{M_b}{M_a} \frac{M_c}{M_b} \frac{M_d}{M_c} \frac{M_e}{M_d}$$

Fuel mass fraction

Segment mass fractions

Mass Fractions

■ Development:

- Use simplified equations to calculate the following for each segment:
 - Fuel mass, Time, Distance
 - Gudmundsson 6.2
- Only consider “Design Mission.”
- Assume Constant C_L cruise
- Don’t forget to use standard atmosphere!

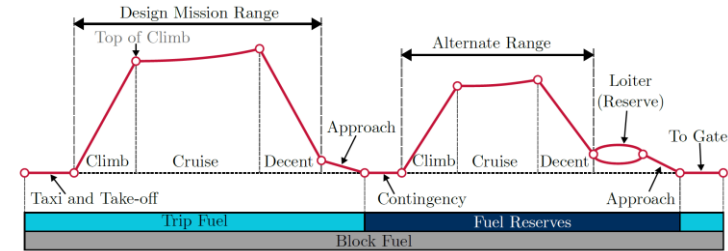
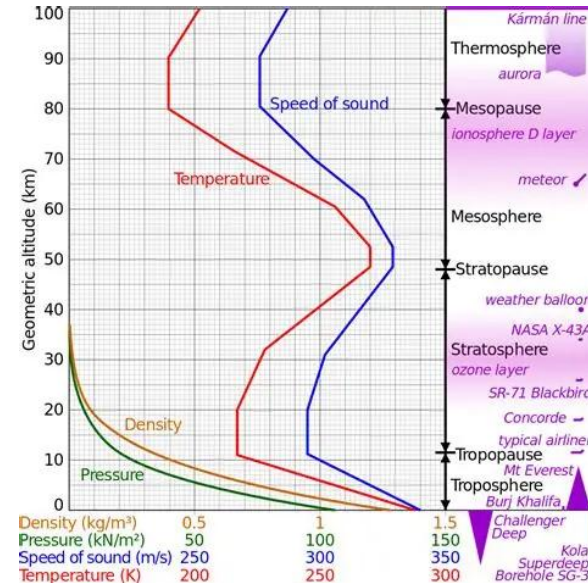


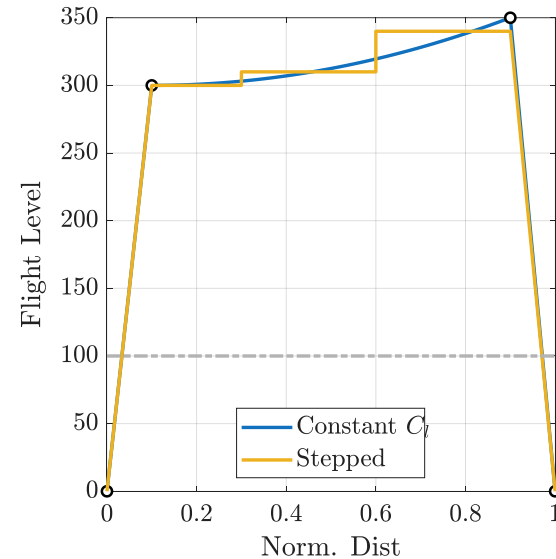
Figure 4: Design mission segment and fuel breakdown



Mass Fractions

▪ Refinement:

- Climb / descent
 - Split each section into mini-segments (e.g., 500ft climb)
 - Different altitude properties
 - Ensure speed above and below 10,000ft is respected
- Cruise
 - Split into mini segments to capture the change in altitude
 - Constant C_L or 1,000ft block climb?
- Missions
 - For block fuel use design mission
 - For climate impact / DOC use all flights
- Use appropriate drag polars in each segment



$$\frac{M_f}{M_a} = \frac{M_a - M_e}{M_a} = 1 - \frac{M_b}{M_a} \frac{M_c}{M_b} \frac{M_d}{M_c} \frac{M_e}{M_d}$$

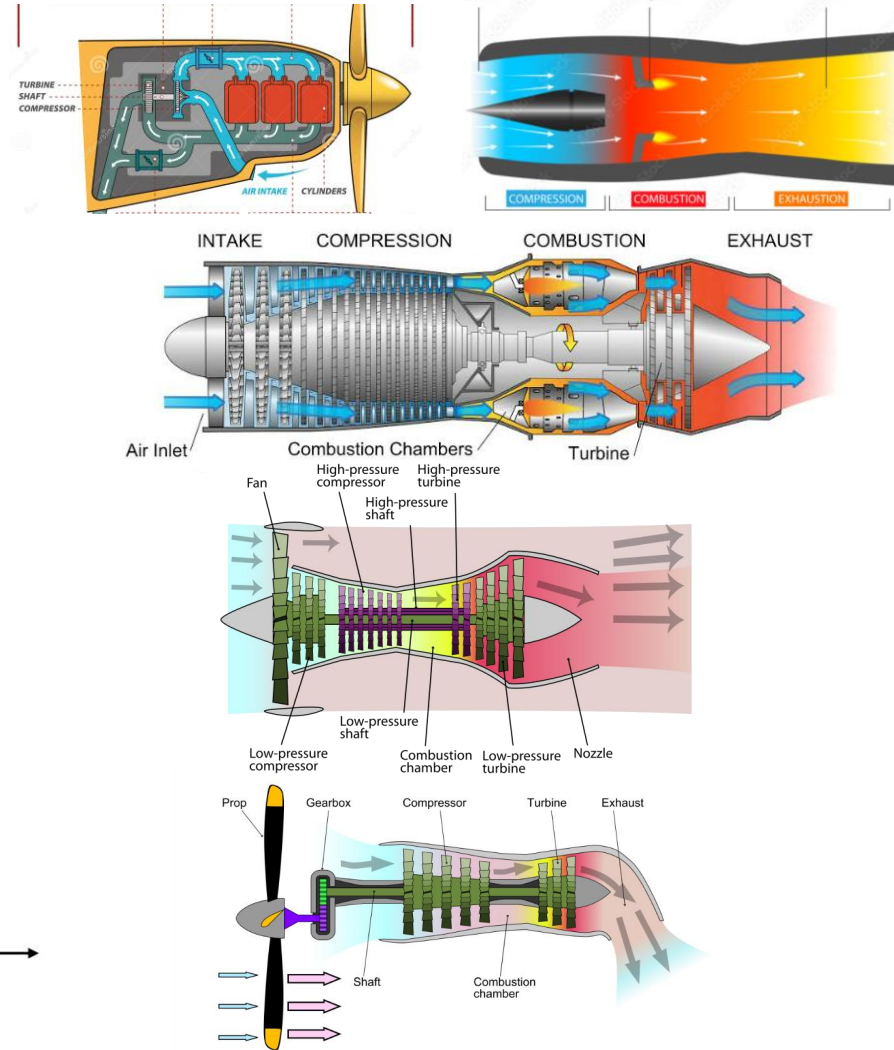
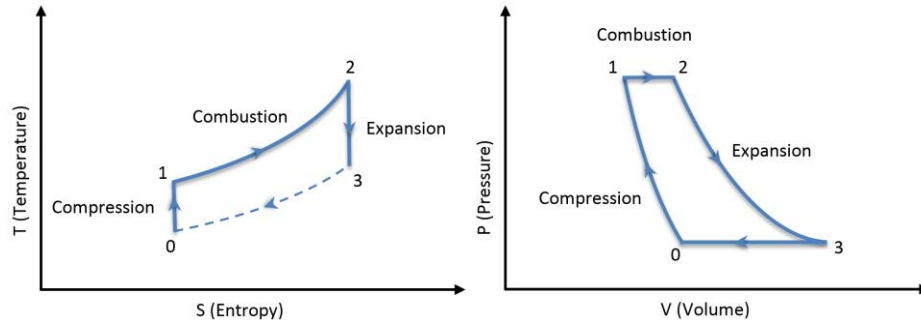
Fuel mass fraction

Segment mass fractions

Propulsion

Choosing a Propulsor

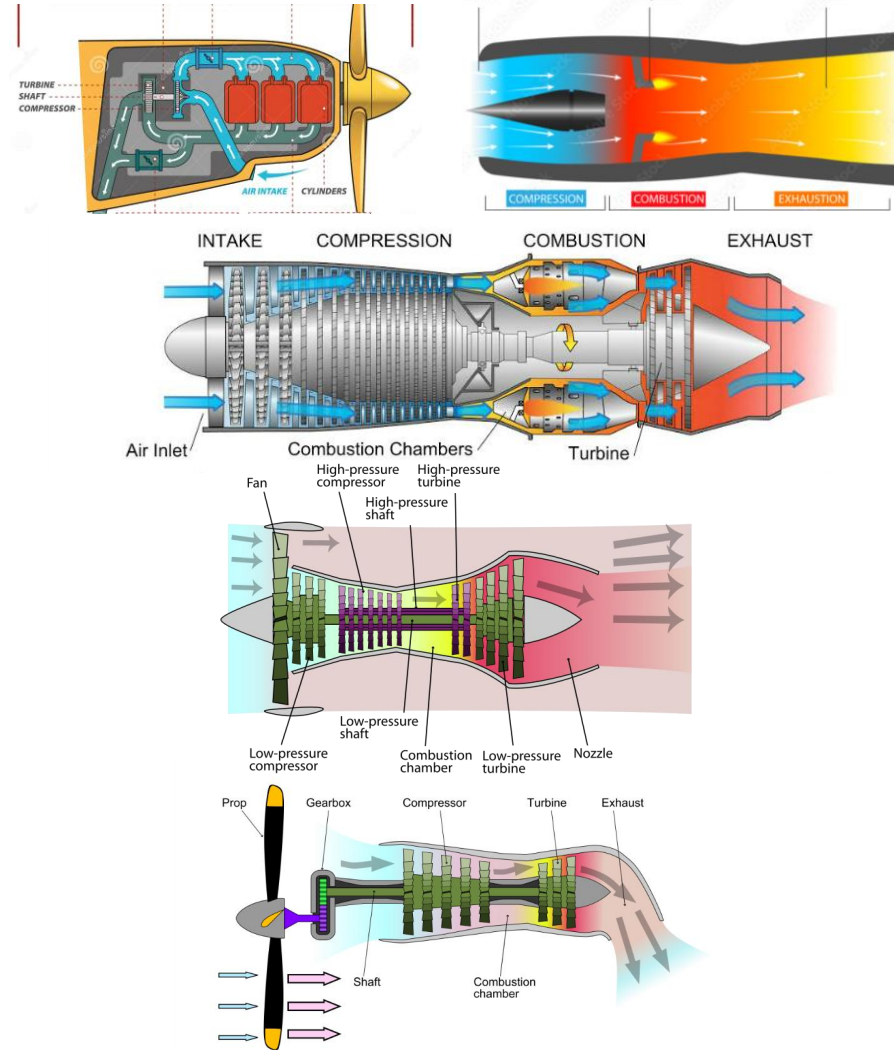
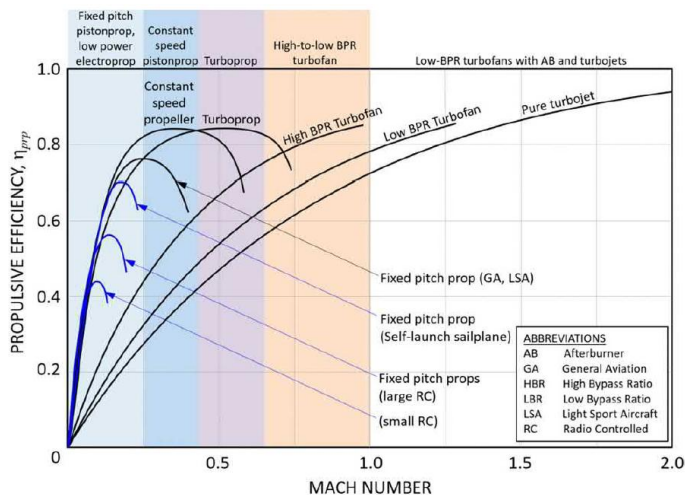
- Most* aircraft engines produce thrust by adding momentum to the oncoming air.
- All work in 3* stages:
 - Compress, combust, exhaust



Choosing a Propulsor

- Selection is mainly driven by:

- Cruise speed
- Efficiency
- Maximum Thrust
- Cost versus Mass...



Turbofan Engines

▪ Key parameters

(~10:1) – Bypass ratio

(~1900K) – Turbine exit temperature

(~50:1) – Overall pressure ratio

– Specific Fuel Consumption

$$BPR = \frac{\dot{m}_{bypass}}{\dot{m}_{core}}$$

$$\pi_c = \frac{p_{tc}}{p_{t0}}$$

$$TSFC = \frac{\dot{m}_f}{F}$$

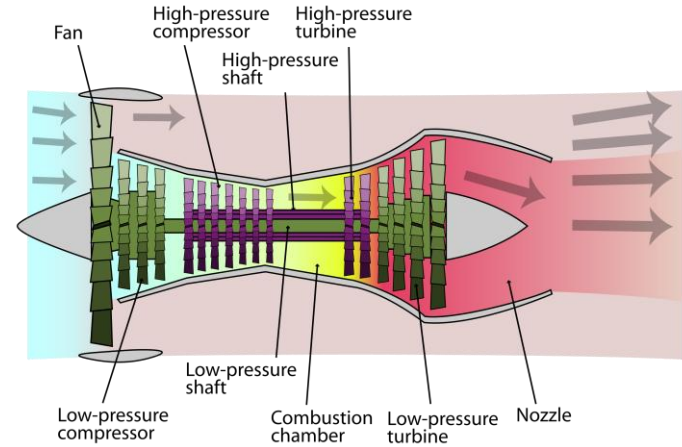


▪ Improved efficiency

➤ Move lots of air slowly rather than a small amount very fast...



▪ Increased frontal area (compared to turbojet)



p_{atm}, T_{atm}

$$p_t = p \left(1 + \frac{\gamma - 1}{2} M_0^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

$$T_t = T \left(1 + \frac{\gamma - 1}{2} M_0^2 \right)$$

$$p_{t3} = p_{t0} \pi_c$$

$$T_{t3} = T_{t0} \pi_c^{\frac{\gamma - 1}{\gamma}}$$

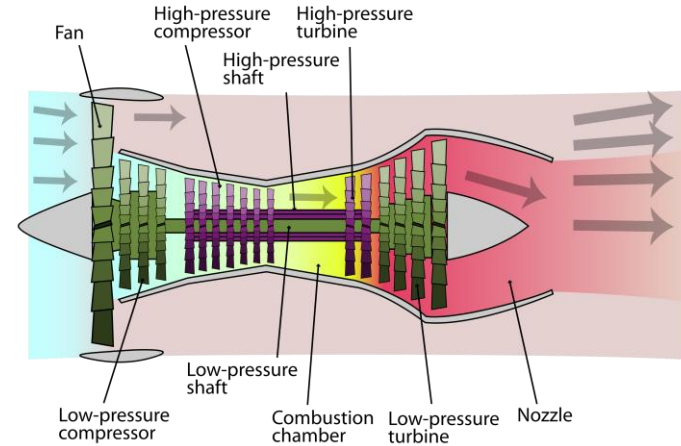
Combustor entry
(isentropic expansion)

Turbofan – Simple Modelling

$$BPR = \frac{\dot{m}_{bypass}}{\dot{m}_{core}} \quad \cancel{\pi_c = \frac{p_{tc}}{p_{t0}}} \quad TSFC = \frac{\dot{m}_f}{F} \quad \cancel{TET}$$

- Use information from the current generation of engines
 - Assume constant pressure ratio and TET
- Use an engine rubberised process to scale dimensions and SFC
- BPR is a defensible hyperparameter
 - Larger diameter fan:
 - Better TSFC
 - Increased mass, drag, and diameter
 - Engine diameter tied to landing gear length...

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Tweak constants for
your engine of choice

Raymer

$$W = 0.084 T^{1.1} e^{(-0.045 \text{ BPR})} \text{ (lb)} = 14.7 T^{1.1} e^{(-0.045 \text{ BPR})} \text{ \{kg\}} \quad (10.4)$$

$$L = 0.185 T^{0.4} M^{0.2} \text{ (ft)} = 0.49 T^{0.4} M^{0.2} \text{ \{m\}} \quad (10.5)$$

$$D = 0.033 T^{0.5} e^{(0.04 \text{ BPR})} \text{ (ft)} = 0.15 T^{0.5} e^{(0.04 \text{ BPR})} \text{ \{m\}} \quad (10.6)$$

$$SFC_{\max T} = 0.67 e^{(-0.12 \text{ BPR})} \text{ (1/hr)} = 19 e^{(-0.12 \text{ BPR})} \text{ \{mg/Ns\}} \quad (10.7)$$

$$T_{\text{cruise}} = 0.60 T^{0.9} e^{(0.02 \text{ BPR})} \text{ (lb)} = 0.35 T^{0.9} e^{(0.02 \text{ BPR})} \text{ \{kN\}} \quad (10.8)$$

$$SFC_{\text{cruise}} = 0.88 e^{(-0.05 \text{ BPR})} \text{ (1/hr)} = 25 e^{(-0.05 \text{ BPR})} \text{ \{mg/Ns\}} \quad (10.9)$$

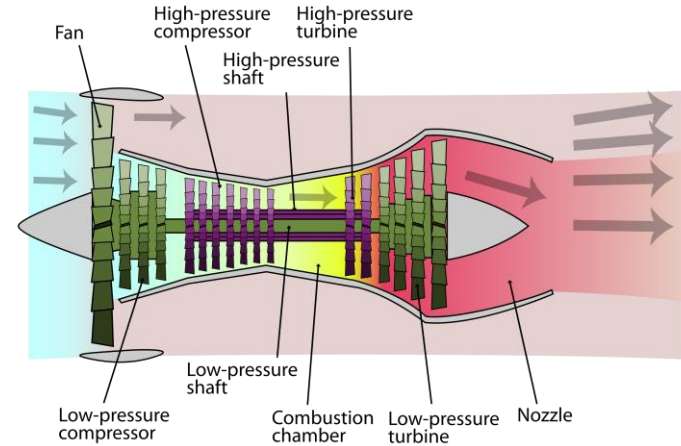
Turbofan – Simple Modelling

$$BPR = \frac{\dot{m}_{bypass}}{\dot{m}_{core}} \quad \cancel{\pi_c = \frac{p_{tc}}{p_{t0}}} \quad TSFC = \frac{\dot{m}_f}{F} \quad \cancel{TET}$$

- Engines are ‘optimised’ for cruise
– Optimal OPR & TET
- Away from this optimum, we can use a simplified approach

$$TSFC = (TSFC_{maxT} + \beta M) * \sqrt{\frac{T}{T_{std}}}$$

- Tune β for your chosen engine
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Tweak constants for
your engine of choice

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Turboprop

- Turbofan with very large BPR...
 - From ~40 to over 100:1...
- Good efficiency at lower Mach numbers
- At higher Mach numbers, the blade tip speed becomes too high, reducing performance
- Propeller engines rated in “power” rather than thrust

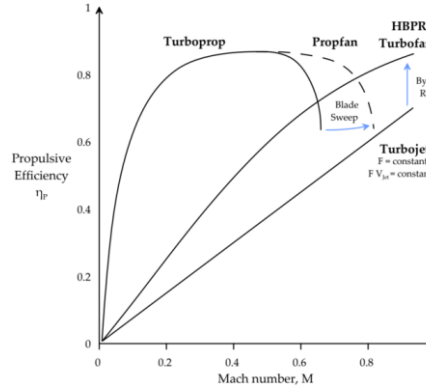
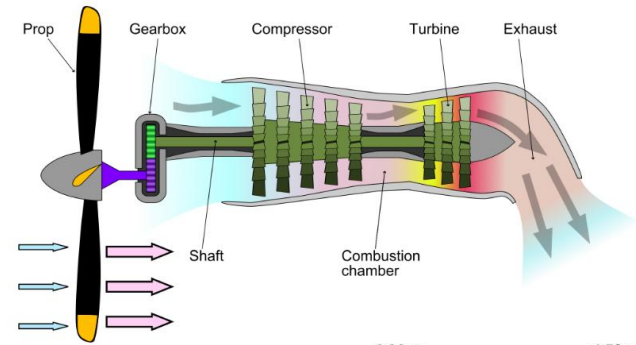
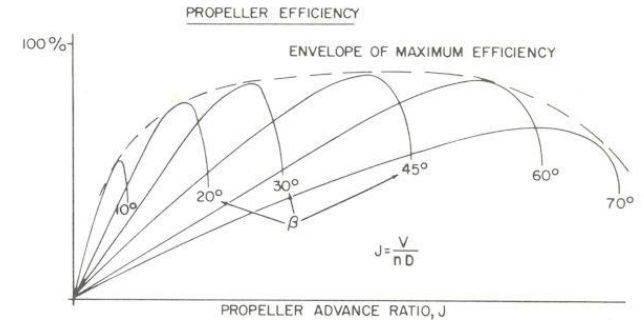
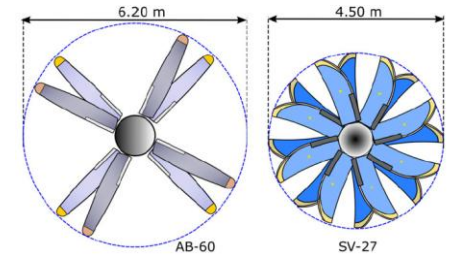


Figure 3. Propulsive efficiency for different engine types.



Turboprop – Simple Modelling

- TSFC function of PSFC, velocity, and propeller efficiency
 - **PSFC**: see current engine data, can assume scales similarly to turbofan
 - They are both turbines!
 - **Propeller efficiency**: tricky to get at the conceptual level.
 - As a first guess, match engine data and drop off at higher Machs

$$TSFC = PSFC \frac{V}{\eta_{prop}}$$

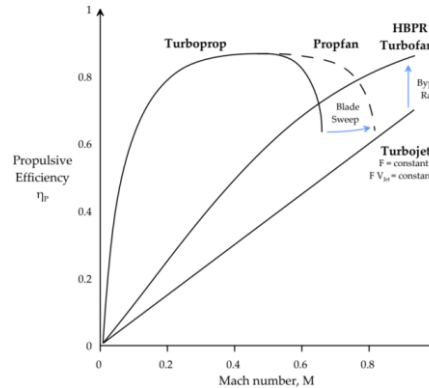
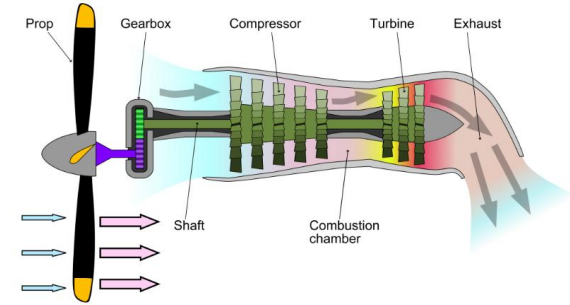
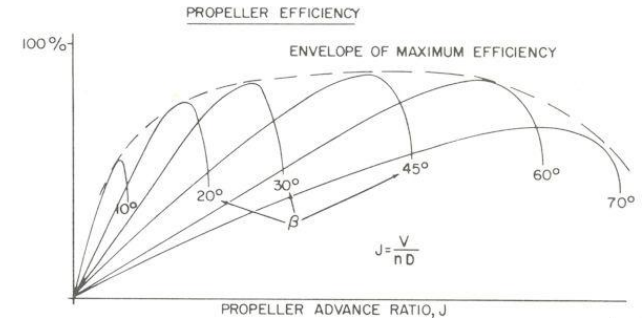


Figure 3. Propulsive efficiency for different engine types.



$$TSFC = PSFC \frac{V}{\eta_{prop}}$$

Help is at Hand

- I've done most of this for you!

Climate Impact

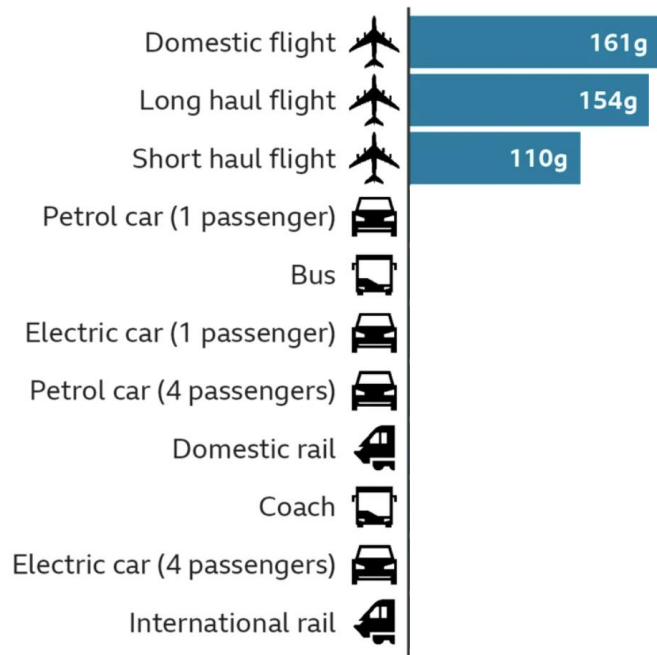
How Bad is Aviation

- Aviation accounts for **~2.5%** of Global CO_2 emissions
- This could rise to **20%** by 2050 if no action is taken*
- But per passenger-per-km, it's efficient, right?
 - Not quite!
- Aviation accounts for **~4%** of all anthropogenic forcing

Emissions from different modes of transport

Emissions per passenger per km travelled

■ CO2 emissions



Note: Figures for average passenger and average vehicle for business travel. Real world emissions may be higher or lower depending on passenger numbers and fuel source, for example. Electric car refers to battery electric vehicle.

Source: DESNZ/Defra Greenhouse Gas Conversion Factors 2023

How Bad is Aviation?

- Passenger numbers are only expected to grow over the coming decades, increasing aviation's yearly emissions.
- But there are opportunities to change the trajectory
 - Ultra-efficient aircraft
 - Sustainable aviation fuel (SAF)
 - Zero-carbon Propulsors
 - Operations
 - ETOPS
 - Less restrictive flight corridors
 - Electric taxi
 - Regulations
 - CORSIA
 - Tax...
 - Placing a cost on emissions

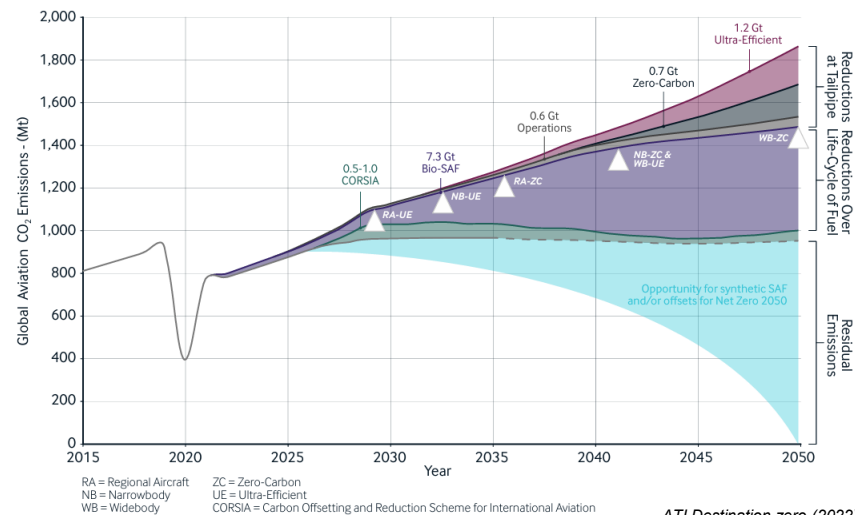
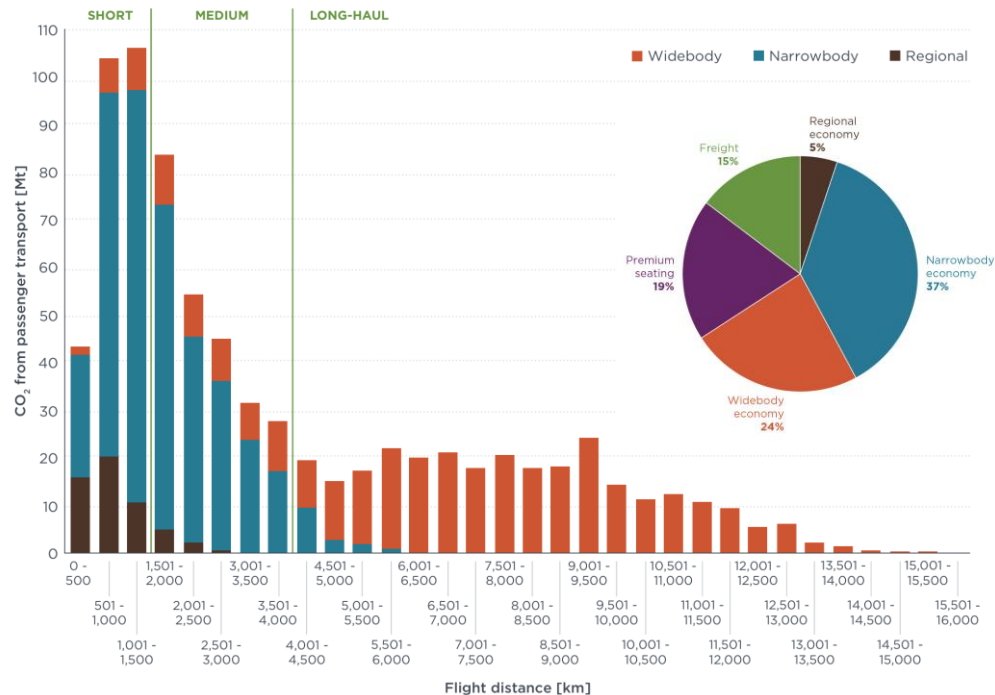


Figure 3 – Impact of future market scenario on Global Aviation CO₂ emissions

How Bad is Aviation?

- **Short-haul flights:**
 - Electric aircraft?
- **Medium haul:**
 - Hydrogen fuel cells?
- **Long-haul:**
 - Will still require ‘combustion’ in 2050. Likely **SAF** (but maybe hydrogen...)
- But SAF and hydrogen are projected to be expensive
 - We are likely in a golden “*price age*” for aviation

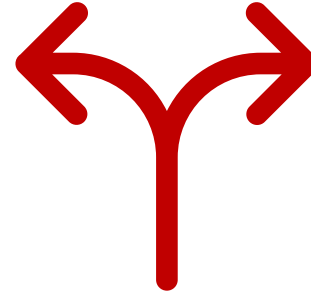


CO₂ Emissions From Commercial Aviation: 2013, 2018, And 2019

Climate Metrics

- Why should we measure climate impact?
 - Policy, public opinion, planning, accountability
- Industry and academia are at a fork in the road, for the first time in aviation, **green** and **cheap** aviation are no longer the same goal...

“green aviation”



“cheap aviation” now

“green aviation”
“cheap aviation”

2000's



1940's

Climate Metrics

- How can we measure climate impact?
 - *We need a quantitative way to estimate the impact of all pollutants emitted by aviation*
 - *Must consider the life-cycle of pollutants*
- **Radiative forcing (RF):**
 - The instantaneous change in the Earth's energy balance caused by a climate driver, expressed as a power per unit area (W/m^2).

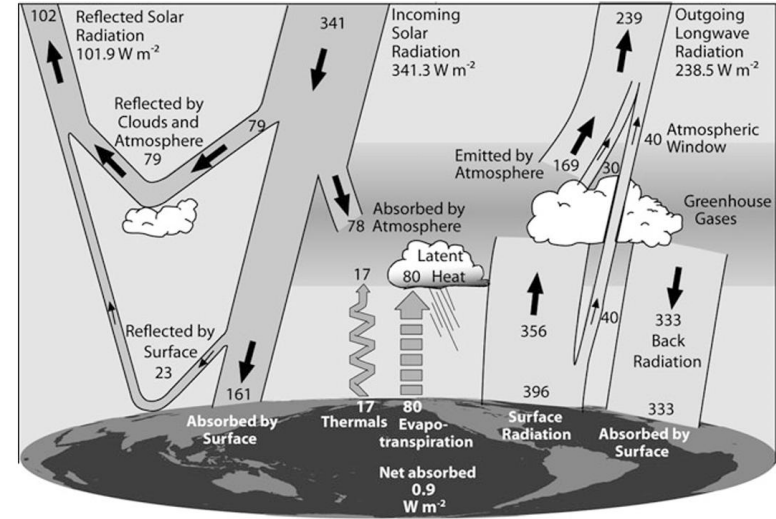
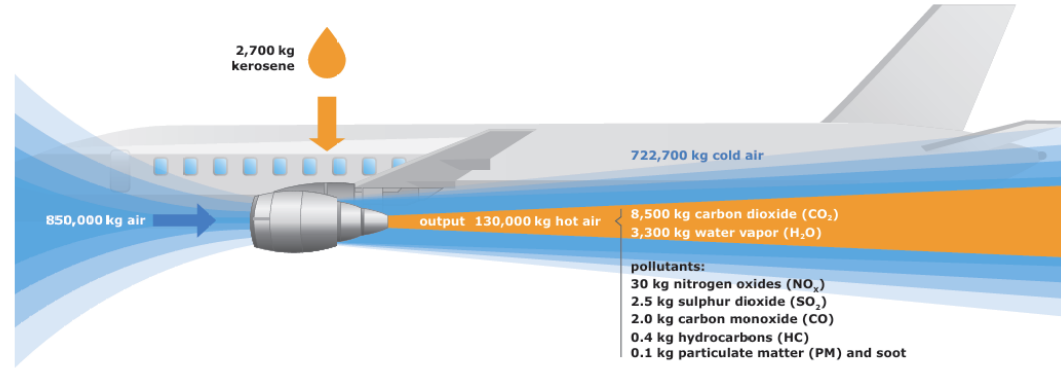


Figure 2.1: Earth's global annual mean energy budget from March 2000 to March 2004. Image by K. E. Trenbeth, National Center for Atmospheric Research, Boulder, Colorado, USA

What's Emitted?

- Burning 1kg of fuel (kerosene) emits:
 - 3.16kg of CO₂ (EI_c)
 - 1.26kg of H₂O (EI_h)
 - $2e^{-4}$ kg of SO₄ (EI_s)
 - $4e^{-5}$ kg of soot (EI_{so})
 - NO_x
 - function of combustor inlet conditions
- It also produces contrails*
 - Aircraft induced cloudiness (EI_{AIC})



European Aviation Environmental Report 2016

$$EI_{NO_x} = 0.0986 \left(\frac{p_3}{101325} \right)^{0.4} \exp \left(\frac{T_3}{194} + \frac{H_0}{53.2} \right)$$

Relative humidity

Converting to RF

- 3.16kg of CO₂ (EI_c)
- 1.26kg of H₂O (EI_h)
- 2e⁻⁴ kg of SO₄ (EI_s)
- 4e⁻⁵ kg of soot (EI_{so})
- NO_x (EI_{so})
- (EI_{AIC})

- Normalised Radiative Forcing:

$$RF^*(t) = \sum_i^{species} Eff_i \frac{RF_i(t)}{RF_{2 \times CO_2}}$$

Still to calculate...

3.7W/m²

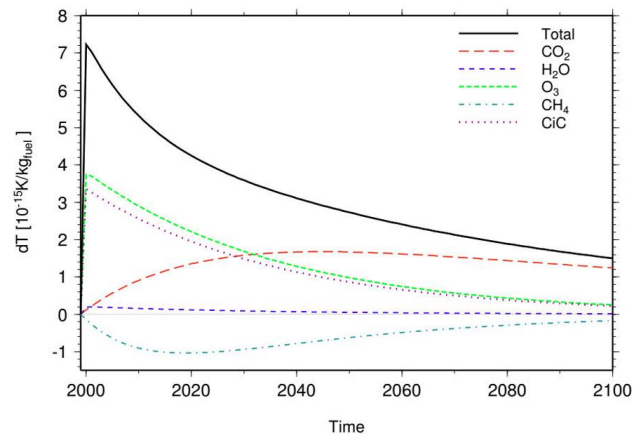
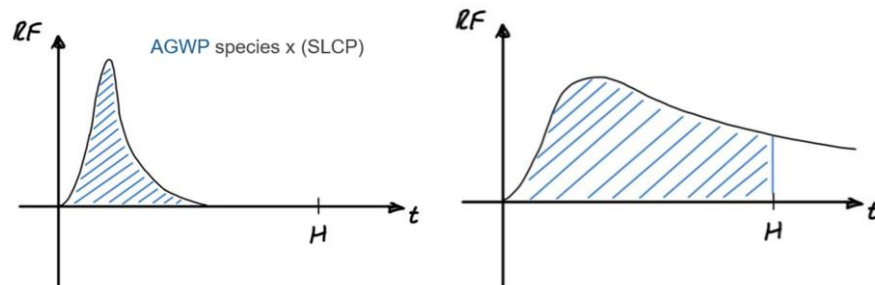
– Equal to one for a doubling in atmospheric carbon dioxide concentration compared to pre-industrial times

	CO ₂	CH ₄	O ₃	H ₂ O	SO ₄	Soot	Contrails
λ_i [K/(W/m ²)]	0.73	0.86	1.00	0.83	0.66	0.51	0.43
Eff_i	1.00	1.18	1.37	1.14	0.90	0.70	0.59

Per kilo, Ozone, methane and water are the worst* “greenhouse gases”

Impact Through Time

- Radiative Forcing indicates **instantaneous forcing**
- But different elements last for different times in the atmosphere!
- What's the best metric to measure climate impact?



Average Temperature Response

- Average increase in **surface temperature** over 'H' years
- Proposed in 2011
- A relatively new consensus that 100 year horizon is an [Home | CLAIM](#) appropriate metric
- One of many metrics
 - Shout out GWP, total change rather than year average

Measured in years*

$$ATR_H = \frac{1}{H} \int_0^H \Delta T(t) dt$$

$$\Delta T(t) = \int_{t_0}^t G_T(t - t') \cdot RF^*(t') dt'$$

$$G_T(t) = \frac{2.246}{36.8} e^{-t/36.8}$$

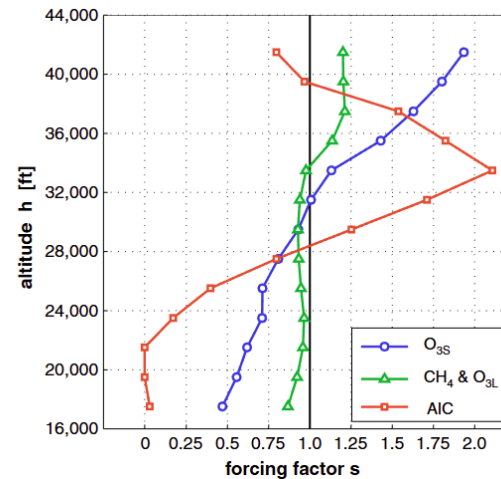


Modelling the thermal response of the Earth to radiative forcing

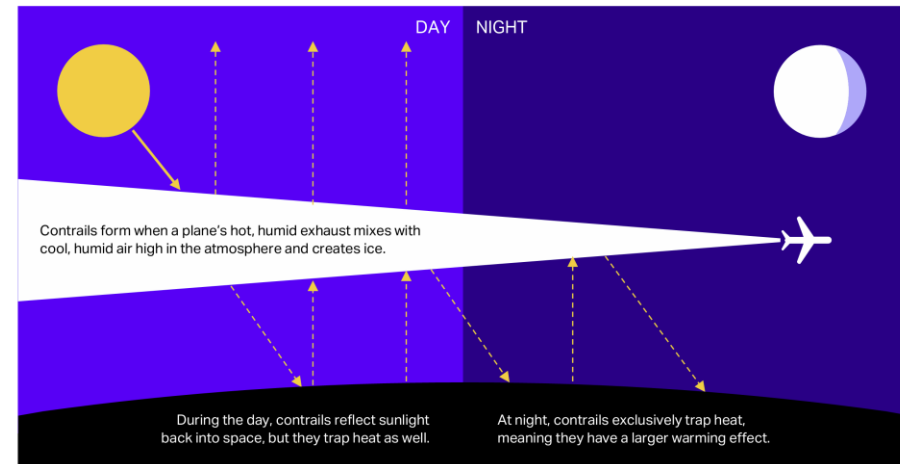
Altitude Dependence

- Long lasting pollutants such as CO_2 show little dependence on emission altitude.
- But other elements have a strong dependence on altitude
- The worst* offender is AIC (aircraft induced cloudiness)*
 - Lots of uncertainty over there actual forcing factor...

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Metric for Comparing Lifetime Average Climate Impact of Aircraft (Dallara 2011)



27 Figure 4 Illustration between day and night contrails. Courtesy of American Airlines

Aviation contrails and their climate effect (2024)

CO₂

- Estimating RF*
- No Altitude dependence

$$\Delta\chi_{\text{CO}_2}(t) = \int_{t_0}^t G_{\chi_{\text{CO}_2}}(t-t') \cdot E_{\text{CO}_2}(t') dt' \text{ with } G_{\chi_{\text{CO}_2}}(t) = \sum_{i=1}^5 \alpha_i \cdot e^{-t/\tau_i}$$

i	1	2	3	4	5
α_i	0.067	0.1135	0.152	0.0970	0.041
τ_i	∞	313.8	79.8	18.8	1.7

$$\text{RF}^*(t) = \frac{1}{\ln 2} \cdot \ln \left(\frac{\chi_{\text{CO}_2,0} + \Delta\chi_{\text{CO}_2}(t)}{\chi_{\text{CO}_2,0}} \right)$$

Background concentration of CO₂ – can assume 380 ppmv*

Water, Soot & Sulfate

- Short-lived – impact considered as average over one year!
 - Again lots of uncertainties built into these values

$$RF_i(t) = \left(\frac{RF_{\text{ref}}}{E_{\text{ref}}} \right)_i \cdot E_i(t) \text{ for } i = \text{Soot, H}_2\text{O, SO}_4$$

$$(RF_{\text{ref}}/E_{\text{ref}})_{\text{H}_2\text{O}} \quad 7.43 \times 10^{-15} \text{ (W/m}^2\text{)}/(\text{kg H}_2\text{O})$$

$$(RF_{\text{ref}}/E_{\text{ref}})_{\text{SO}_4} \quad -1.00 \times 10^{-10} \text{ (W/m}^2\text{)}/(\text{kg SO}_4)$$

$$(RF_{\text{ref}}/E_{\text{ref}})_{\text{soot}} \quad 5.00 \times 10^{-10} \text{ (W/m}^2\text{)}/(\text{kg soot})$$

Aviation Induced Cloudiness

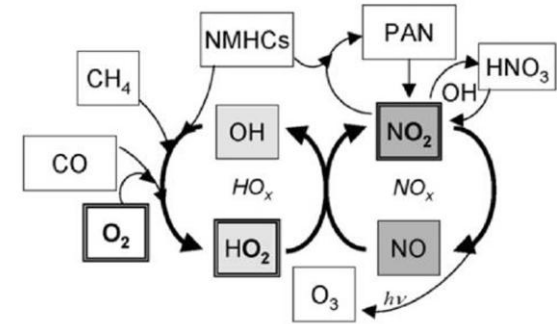
- In this simplified model, you can use altitude dependence on slide 28 (10.2514/1.J050763)
- L is the stage length flown per year (in miles...)
- If you're interested, other AIC modelling methods exist...

$$\text{RF}_{\text{AIC}}(t, h) = s_{\text{AIC}}(h) \left(\frac{\text{RF}_{\text{ref}}}{L_{\text{ref}}} \right)_{\text{AIC}} L(t)$$

$(\text{RF}_{\text{ref}}/L_{\text{ref}})_{\text{AIC}} \quad 2.21 \times 10^{-12} \text{ (W/m}^2\text{)}/\text{n mile}$

NO_x (the fun one)

- NO_x is not a greenhouse gas... but it's quote reactive...
- ✓ ■ It depletes methane and Ozone in the long-term
– s(h) from slide 28
- ✗ ■ It generates ozone in the short term



$$RF_i(t, h) = s_i(h) \int_{t_0}^t G_i(t - t') \cdot E_{NO_x}(t') dt'$$

with $G_i(t) = A_i \cdot e^{-t/\tau_n}$ for $i = CH_4, O_{3L}$

$$A_{CH_4} = -5.16 \times 10^{-13} (W/m^2)$$

$$A_{O_{3L}} = -5.16 \times 10^{-13} (W/m^2)$$

$$\tau_n = 12 \text{ yrs}$$

$$RF_{NO_x-O_{3S}}(t, h) = s_{NO_x-O_{3S}}(h) \cdot \left(\frac{RF_{ref}}{E_{ref}} \right)_{NO_x-O_{3S}} \cdot E_{NO_x}(t)$$

$$1.01 \times 10^{-11} (W/m^2)$$

Over to You!

▪ Development:

- Calc RF^* for each species during cruise
- Calc ATR_{100} (e.g. 100 year horizon)

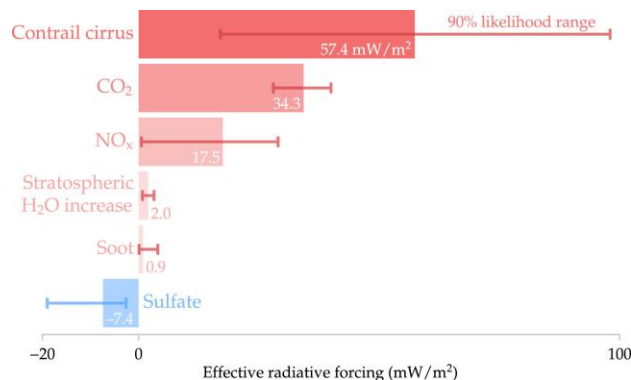
▪ Refinement:

- Complete climate analysis for the entire mission (altitude dependence)

$$RF^*(t) = \sum_i^{species} Eff_i \frac{RF_i(t)}{RF_{2 \times CO_2}}$$

$$\Delta T(t) = \int_{t_0}^t G_T(t - t') RF^*(t') dt'$$

$$ATR_{100} = \frac{1}{100} \int_0^{100} \Delta T(t) dt$$



Main References:

Metric for Comparing Lifetime Average Climate Impact of Aircraft (Dallara 2011) <https://doi.org/10.2514/1.J050763>
Airplane Design Optimization for Minimal Global Warming Impact (Proesmans 2022) <https://doi.org/10.2514/1.C036529>

Conclusions

Conclusions

- Mission Analysis
 - Refresher on required mass fractions
- Propulsion
 - Laid out the “rubberization” of turbofan and turbo engines
- Climate Impact
 - Laid out framework to estimate climate impact
- Sprints
 - Laid out expected development in each sprint
- SAF
 - Life cycle emissions analysis; I will update you later...

Questions?